

Desmos Mastery - Day 6

10 original questions | Calculator permitted

Complete the questions before turning to the answer explanations. Mark any guess, even if it turns out to be correct. For math, show your setup and check the requested quantity. For reading, identify the exact supporting detail; for grammar, identify the clause or subject.

These questions were authored for this workbook from the lesson topics. They are not official SAT questions or copies of the course's paid practice.

Regression Part 2 | 10 questions

[Lesson video: Regression Part 2](#)

1. The equation below has two positive solutions a . What is their product?

$$a + \frac{9}{a} = 10$$

Answer: _____

2. The identity below holds for all real x . What is $p + q$?

$$(x + 4)^2 - 9 = x^2 + px + q$$

Answer: _____

3. The graphs below intersect at a point whose y -coordinate is 13. What is k ?

$$y = 2x + 3, \quad y = kx - 12$$

Answer: _____

4. A data set contains $(0, 5)$, $(1, 8)$, $(2, 17)$, and $(3, 34)$. The point $(3, 34)$ is removed as an outlier. The remaining points lie exactly on $f(x) = ax^2 + c$. What is a/c ?

Answer: _____

5. For all real x in the domain, the rational identity below holds. What is k ?

$$\frac{x^2 - 4}{x - 2} = x + k \quad (x \neq 2)$$

Answer: _____

6. The equations below have a common solution. What is $8^x/2^y$?

$$3x - y = 4$$

$$x + y = 8$$

Answer: _____

7. For $f(t) = t^2 + 2t$, a positive number u satisfies $f(u) = 35$. What is u ?

Answer: _____

8. A student uses a numerical fit to solve a parameter equation and obtains $a = -4$. The problem states $a > 0$. What is the correct conclusion?

- A) The negative sign can be ignored.
- B) The original equation has no solutions of any kind.
- C) This output is inadmissible; search for an allowed root and verify it.
- D) Every fit returns the smallest positive root.

9. The identity below holds for every real x . What is r ?

$$(x - 2)(x - r) = x^2 - 8x + s$$

Answer: _____

10. A quadratic has roots whose sum is 12 and whose product is 20. Its leading coefficient is 3. What is its constant term?

Answer: _____

Answers & explanations

Count correct answers, but also review correct guesses. Redo missed questions after a day without looking at the solution.

1: 9 2: 15 3: 5 4: 0.6 5: 2 6: 16 7: 5 8: C 9: 6 10: 60

Why each answer works

1. Answer: 9. Rearrange to $a^2 - 10a + 9 = 0 = (a - 1)(a - 9)$. The solutions are 1 and 9, whose product is 9. One numerical regression result alone would miss the second solution.

2. Answer: 15. Expand to $x^2 + 8x + 16 - 9 = x^2 + 8x + 7$. Thus $p = 8$ and $q = 7$, so $p + q = 15$.

3. Answer: 5. The known y-value gives $13 = 2x + 3$, so $x = 5$. Then $13 = 5k - 12$ gives $k = 5$. Fix the given y-value rather than fitting it as an additional unknown.

4. Answer: 0.6. The input 0 gives $c = 5$. The input 1 gives $a + 5 = 8$, so $a = 3$. The input 2 verifies $4(3) + 5 = 17$. Thus $a/c = 3/5 = 0.6$. Removing a point changes the fitted model; distinguish a coefficient from an output value.

5. Answer: 2. Factor the numerator as $(x - 2)(x + 2)$. The simplified expression is $x + 2$ for $x \neq 2$, so $k = 2$. Preserve the excluded input.

6. Answer: 16. Write the ratio as $2^{3x - y}$. The first equation fixes the exponent at 4, so the ratio is 16 regardless of the individual coordinates.

7. Answer: 5. $u^2 + 2u - 35 = 0$ factors as $(u + 7)(u - 5)$. Positivity selects $u = 5$. Defining a function and fitting its input does not remove the need to check the input restriction.

8. Answer: C. A root can satisfy an equation while violating the stated domain. Numerical routines can return different roots; they do not certify completeness or admissibility.

9. Answer: 6. The x-coefficient on the left is $-(2 + r)$. On the right it is -8 , so $r = 6$. The constant s is 12.

10. Answer: 60. The monic polynomial is $x^2 - 12x + 20$. Multiplying by the leading coefficient 3 gives constant term 60. This combines root information with model scaling.

For each missed or guessed question, record the skill, the condition you overlooked, and one specific action for next time.